EGX Stock Market- Journey Report

EGX Stock Market — Data Acquisition Report

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Dataset: Egyptian Exchange (EGX) — 32 Shariah-Index Companies
Period: June 2025 – December 2025 | 4,640 trading records
Notebooks: `api_fetcher.ipynb` · `Bs4.ipynb` · `SeleniumProject.ipynb` · `cleaning_analysis.ipynb`

Abstract :

This project focuses on the acquisition and preparation of financial data for Shariah-compliant companies listed on the Egyptian Exchange (EGX). The dataset includes 32 companies over the period from June 2025 to December 2025, resulting in 4,640 trading records stored in `raw.csv`.

The data was collected from multiple sources, including financial APIs, web scraping, and dynamic scraping techniques. The dataset was then organized and processed using the Pandas library. NetworkX was used for graph-based analysis, including measures such as betweenness centrality, while Matplotlib was used for data visualization.

The main objective of the project is to build a solid analytical foundation of the Egyptian market in order to support investment decision-making by analyzing sectors, companies, and industries through financial data analysis, including stock performance evaluation and market behavior study.

Overall, this project demonstrates practical experience in data acquisition, integration, preprocessing, and basic financial network analysis within a real-world market context.

Introduction

Background

The Egyptian Exchange (EGX) is one of the key financial markets in the Middle East and North Africa region, where a wide range of companies are listed and traded daily. Stock market analysis plays a crucial role in understanding market behavior, identifying trends, and supporting informed investment decisions. By analyzing historical trading data, it is possible to gain insights into price movements, volatility, and relationships between different companies and sectors.

Objective

The main objective of this project is to analyze Shariah-compliant stocks listed on the Egyptian Exchange (EGX) using data science techniques. The project aims to extract meaningful insights from financial data in order to understand stock performance and market behavior. This includes evaluating trends across companies and sectors and applying analytical methods to support better financial interpretation of the Egyptian market.

Dataset Description

Data Scope

This project uses two complementary datasets related to the Egyptian Exchange (EGX) to support financial analysis.
The **primary dataset** consists of 32 Shariah-compliant companies listed on the EGX. It covers the period from June 2025 to December 2025, resulting in approximately 4,640 trading records stored in `raw.csv`. This dataset captures daily stock market behavior across multiple companies and sectors.
The **secondary dataset** was collected from *African Markets* using Selenium automation. It includes EGX30 index historical data, weekly market summaries, and company metadata such as sector classification (e.g., healthcare, financial services, industrial). This dataset was used to provide broader macroeconomic and sector-level context to complement individual stock analysis which stored in `stocks_data.csv`.

Both datasets together gives a great overall and overview of EGX market.

Features

The datasets contains a set of financial indicators used to analyze stock performance, risk, and market behavior across Egyptian Exchange (EGX) companies:

- **Open Price**: The first trading price of a stock during a given day
- **Close Price**: The final trading price at market close
- **High / Low**: The maximum and minimum prices recorded during the trading session
- **Volume**: The number of shares traded in a single day
- **Returns**: Percentage change in price used to measure performance over time
- **Volatility**: Statistical measure of price fluctuations used to evaluate risk levels
- **Market Capitalization**: Total market value of a company’s outstanding shares
- **Sector Classification**: Industry grouping used for macro-level analysis and comparison

Data Sources

The dataset was constructed using multiple data collection techniques to ensure completeness, reliability, and analytical depth:

- **Financial API (yfinance):** Used to extract historical OHLCV (Open, High, Low, Close, Volume) data for 32 EGX-listed companies over the selected time period
- **Web Scraping (BeautifulSoup):** Initially attempted for extracting static financial data from web sources; however, it was limited due to JavaScript-rendered content on target websites
- **Dynamic Web Scraping (Selenium):** Used to extract structured data from interactive websites such as African Markets, including sector classifications, market summaries, and additional company-level indicators

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Data Collection

Issues Faced

During the initial stage of data collection, multiple approaches were tested to retrieve a reliable list of EGX-listed companies and their financial data. The first attempt used `SerpApi` using **Google Finance engine** to extract stock listings and company information. However, this method did not work as expected because there is no data for EGX companies. Because of this, we looked for another solution and found the **Yahoo Finance API (** `yfinance` **)**.

We also faced another problem when using the Yahoo Finance API (`yfinance`). We needed to know the correct stock ticker symbols for EGX companies, including the `.CA` suffix. However, we did not initially know the correct ticker names or formats for all companies.

Because of this, we had to search for and collect the correct ticker symbols manually from different sources to ensure the API would work properly and return valid data.

After solving these issues, the Yahoo Finance API worked well and allowed us to collect structured and complete historical stock data for all selected companies.

To address this issue, alternative data sources were adopted, including:

- Financial API extraction using `yfinance` for reliable historical stock data
- Web scraping techniques using BeautifulSoup for static content
- Dynamic scraping using Selenium for JavaScript-rendered financial websites

This multi-source strategy ensured complete coverage of the required EGX companies and improved the reliability of the final dataset.

Data Cleaning & Preprocessing

Steps

During the preprocessing stage, we found that some columns in the primary dataset — **'Capital Gains', 'Dividends', and 'Stock Splits'** — contained only zero values and did not provide useful information for analysis.
Next, column names were cleaned by removing extra spaces to ensure consistency across the dataset.
The **Date** column was then converted into a proper datetime format. Time zone information was also removed.
Finally, the dataset was checked for missing values using `isna().sum()`, which confirmed the presence or absence of null values in each column.

In the secondary dataset was cleaned to ensure consistency and improve data quality before analysis.
First, only valid and relevant sectors were kept. A predefined list of sectors (such as Financials, Industrials, Technology, etc.) was used to filter the dataset. Any rows that did not match these sectors were removed.
Next, rows with missing values in important columns like **Company** and **Sector** were dropped to ensure complete and reliable records.
After that, numeric columns such as **Price, Market Capitalization (M.Cap), 1D change, and YTD performance** were originally stored as text. These values were cleaned by removing symbols like `%`, `+`, and commas, and then converted into proper numeric format for analysis.

During preprocessing, it was necessary to align company names across both datasets to ensure a successful merge and correct sector assignment.
Many companies appeared in the two datasets under slightly different names. For example, a company might appear as `Fawry` in one dataset and as `Fawry for Banking Technology and Electronic Payment` in the other. If these names are not matched correctly, the datasets cannot be merged properly and sector information will be missing.
To solve this issue, we first identified the unmatched company names between both datasets. Then, we created a mapping dictionary to standardise the names so that each company has a consistent label in both datasets.
After applying this mapping, all company names in the primary dataset were aligned with the secondary dataset, allowing successful merging and correct assignment of sector information

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Processing the Data

After cleaning and preparing the datasets, several financial features were calculated to support further analysis of stock performance, risk, and portfolio behavior.

Formulas Used

The following formulas were applied:

- **Price Range**
  Range = (High − Low) / Close
- **Period Return (%)**
  Period Return = [(Close − Open) / Open] × 100
- **Price vs 365-Day High/Low (%)**
  Price vs 365 High/Low = [(Close − $High/Low_{365}$) / $High/Low_{365}$] × 100
- **Portfolio Value (Cumulative Return)**
  Portfolio Value = Start Value × $\prod(1 + r_t)$
- **Total Gain**
  Total Gain = Portfolio Value − Start Value

Purpose of These Features

These calculated features were used to:

- Measure daily stock performance
- Analyze price movement and trends
- Evaluate risk through volatility and return behavior
- Simulate portfolio growth over time
- Compare companies and sectors on a consistent basis

Network Analysis

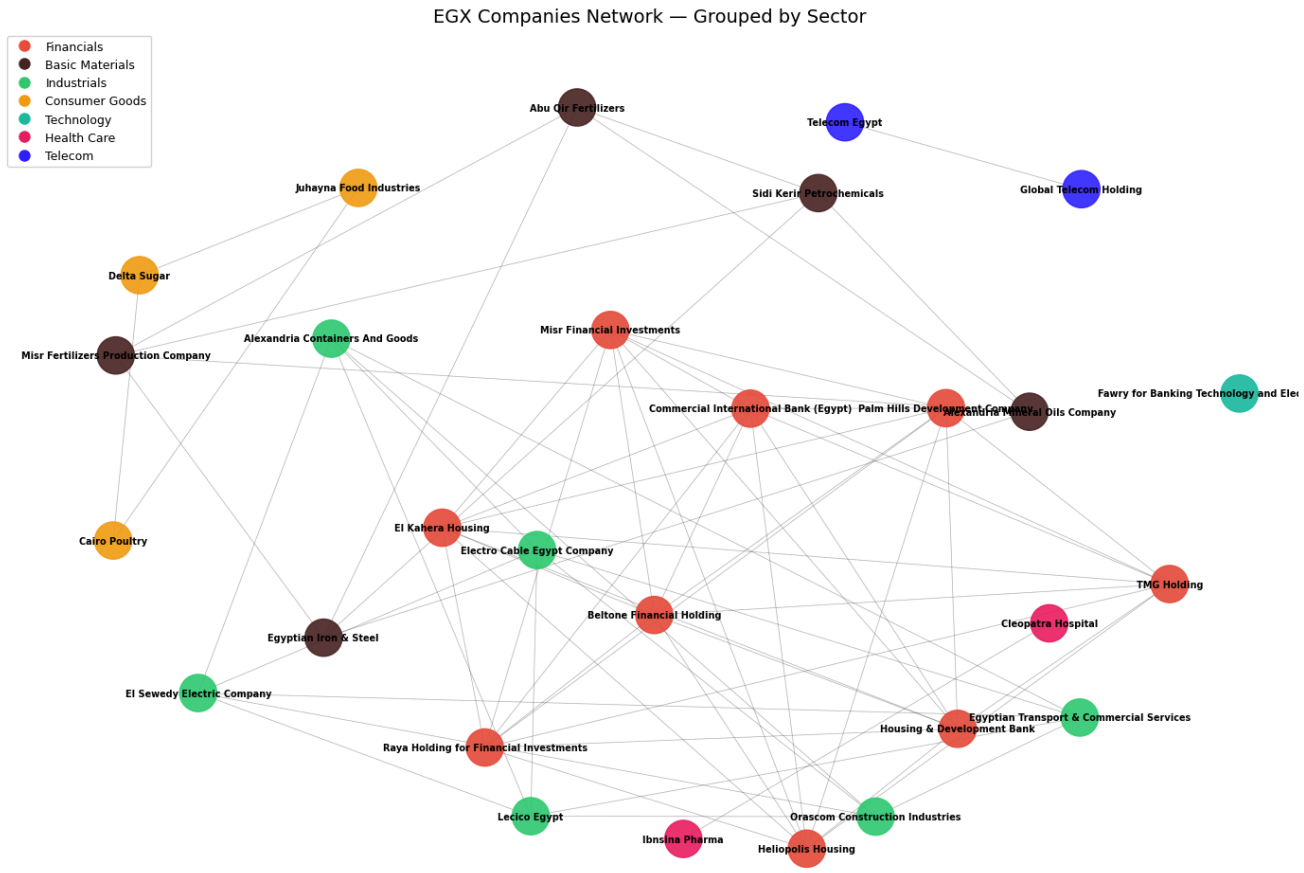
This section analyzes the relationships between companies in the Egyptian Exchange (EGX) using a network-based approach. The goal is to understand how companies are grouped based on their sectors.

Graph Construction

A network graph was built using sector information:

- Each node represents a company
- Companies are connected if they belong to the same sector
- This forms clusters of companies within each industry

Visualization



Degree Centrality

Degree centrality measures how connected each company is within its sector group.

Top companies by degree centrality:

- Beltone Financial Holding
- Commercial International Bank (Egypt)
- El Kahera Housing
- Heliopolis Housing
- Housing & Development Bank

All these companies have similar values because they belong to similarly sized sector groups.

Betweenness Centrality

Betweenness centrality measures whether a company connects different groups.

Top companies:

- Abu Qir Fertilizers
- Alexandria Mineral Oils Company
- Egyptian Iron & Steel
- Misr Fertilizers Production Company
- Sidi Kerir Petrochemicals

All values are zero because the network only connects companies within the same sector, so there are no cross-sector links.

Summary

The network shows clear clustering by sector. Companies in the same industry are grouped together, but there are no connections between different sectors in this model.

Sector Analysis & Portfolio Simulation

Sector Performance

Sector performance was evaluated using:

- Average Daily Return (primary dataset)
- YTD Return (secondary dataset)

Sector Statistics

For each sector:

- Average Daily Return (performance)
- Volatility (risk)
- Total Volume (activity)
- Market Capitalization (size)

Portfolio Simulation

- Initial investment: 10,000 EGP
- Returns averaged per sector
- Compounded over time

Portfolio Value = Initial × ∏(1 + Daily Return)

Sector Portfolio Growth

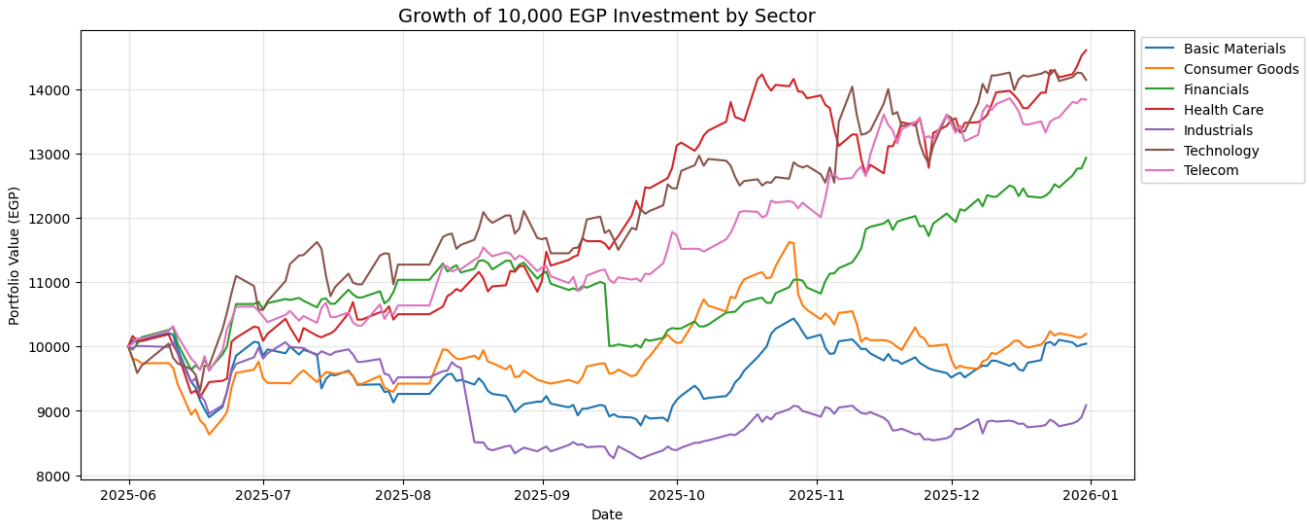


Figure 1: Growth of a 10,000 EGP investment across sectors.

Insight

Overall, the portfolio values show an increasing trend over time, indicating positive returns across most sectors.

The Healthcare, Technology, and Telecom sectors achieved the highest growth, showing the best performance during the study period. In contrast, the Industrials, Consumer Goods, and Basic Materials sectors showed slower growth and lower final portfolio values.

Company Risk Analysis (Annualised Volatility)

To measure risk across companies, daily returns were used to compute both annualised return and annualised volatility.

- Annualised Return was calculated by scaling the mean daily return:
  Ann_Return = Mean_Daily_Return × 252
- Annualised Volatility was calculated using standard deviation:
  Ann_Volatility = Std_Daily_Return × √252

Where 252 represents the average trading days per year.

Each company was also mapped to its corresponding sector using the sector mapping created earlier.

Volatility Visualization

A bar chart was plotted to compare annualised volatility across companies.

- Companies were sorted in descending order of volatility
- Higher bars indicate higher risk
- Lower bars indicate more stable stocks

Figure

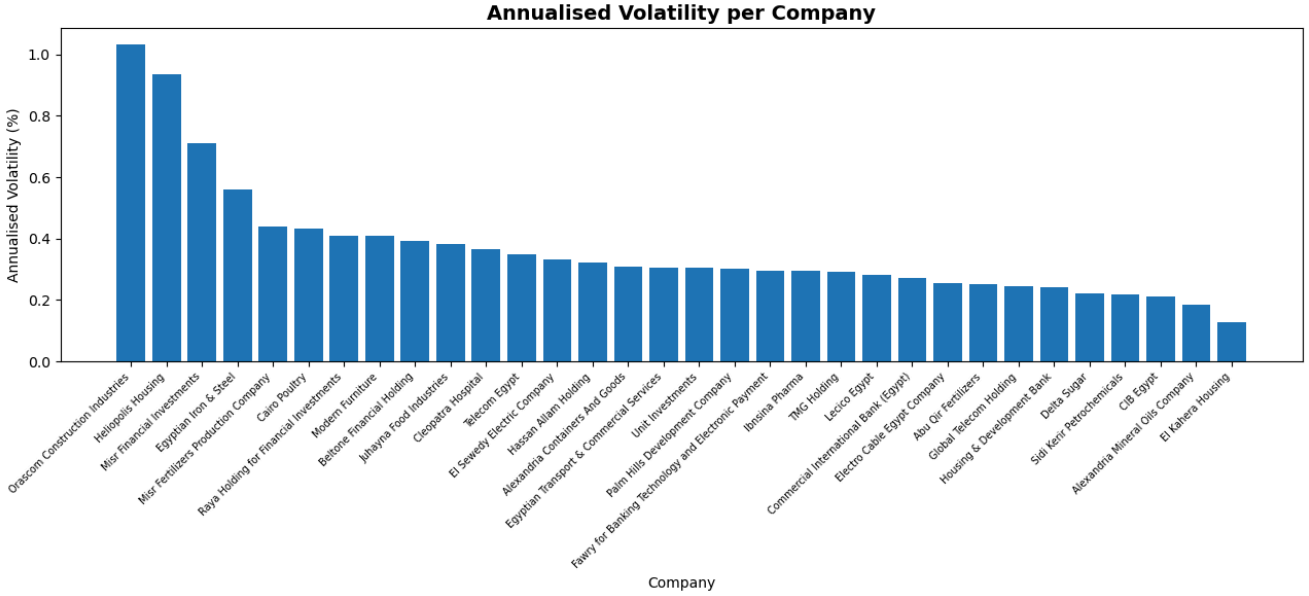


Figure: Annualised volatility comparison across companies, showing differences in risk levels between stocks.

Insight

The results show clear differences in risk across companies. Higher volatility indicates higher risk, while lower volatility indicates more stable stocks.

Most risky companies (highest volatility):

- Orascom Construction Industries
- Heliopolis Housing
- Misr Financial Investments
- Egyptian Iron & Steel
- Misr Fertilizers Production Company

Safest companies (lowest volatility):

- El Kahera Housing
- Sidi Kerir Petrochemicals

- Alexandria Mineral Oils Company
- Housing & Development Bank
- Global Telecom Holding

Overall, risk levels vary significantly across companies, showing different stability patterns in the Egyptian market.

Rolling Volatility Analysis (Top 5 Market Cap Companies)

This section analyzes how risk changes over time for the five largest companies by market capitalization using a 30-day rolling volatility measure.

Rolling volatility captures short-term fluctuations in stock returns, making it useful for identifying periods of instability.

Method

Volatility was calculated using a 30-day rolling window and annualised using:

Rolling Volatility = std(Daily_Return, 30-day window) × √252 × 100

This allows comparison of risk levels across companies on a consistent annual scale.

Results Summary

Commercial International Bank (Egypt)

- Mean Volatility: 25.87%
- Max Volatility: 37.05%
- Min Volatility: 16.43%

Misr Fertilizers Production Company

- Mean Volatility: 37.68%
- Max Volatility: 92.88%
- Min Volatility: 11.18%

TMG Holding

- Mean Volatility: 29.05%
- Max Volatility: 41.89%

- Min Volatility: 15.81%

El Sewedy Electric Company

- Mean Volatility: 24.97%
- Max Volatility: 65.31%
- Min Volatility: 9.28%

Abu Qir Fertilizers

- Mean Volatility: 25.21%
- Max Volatility: 39.71%
- Min Volatility: 11.68%

Visualization

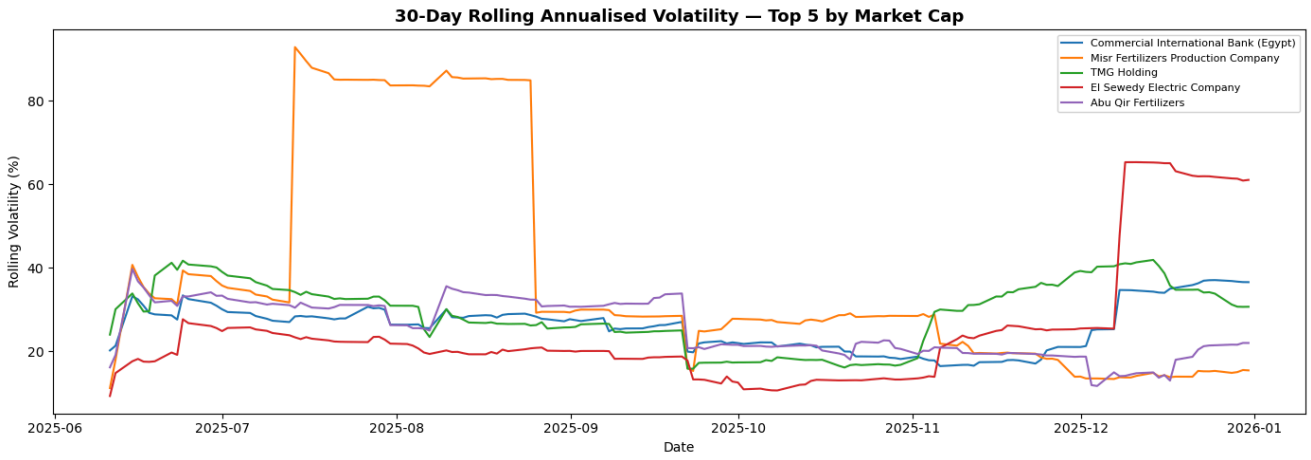


Figure: 30-day rolling annualised volatility for the top 5 companies by market capitalization.

Insight

Misr Fertilizers is the most risky company, showing very large fluctuations in price. This means its stock value can change quickly and is less stable compared to others.

In contrast, Commercial International Bank (CIB) and El Sewedy Electric are more stable, with smoother price movements and lower risk.

Overall, even large companies in the Egyptian market behave differently. Some are stable and safer, while others are more volatile depending on market conditions.
Overall, even large-cap companies in the Egyptian market exhibit varying levels of risk depending on market conditions.

Investment Insight (Based on Analysis)

Based on the analysis of returns, volatility, and sector performance, some sectors show stronger overall performance compared to others.

Strong Performing Sectors (Higher Growth Potential)

- Healthcare
- Technology
- Telecom

These sectors showed higher growth and relatively strong performance during the study period. They are generally supported by stable demand (Healthcare), digital expansion (Technology), and continuous usage needs (Telecom).

Moderate Performance Sectors

- Financials
- Industrials

These sectors showed mixed results, with both stable companies and more volatile ones. Their performance depends heavily on economic conditions and company size.

Weaker Performing Sectors (Lower Growth in This Period)

- Basic Materials
- Consumer Goods

These sectors showed slower growth and weaker overall performance during the study period. They are more affected by production costs and economic fluctuations.

Company-Level Insight (From Data Results)

- Lower risk and relatively stable companies include:
  Commercial International Bank (CIB), El Sewedy Electric, Abu Qir Fertilizers
- Higher risk companies include:
- Misr Fertilizers Production Company, Orascom Construction Industries, Heliopolis Housing

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Conclusion

This project presented a full data science pipeline applied to the Egyptian Exchange (EGX), focusing on 32 Shariah-compliant companies over the period from June 2025 to December 2025. The work included data collection, preprocessing, feature engineering, financial analysis, and network modeling.

The data was collected from multiple sources, including financial APIs and web scraping tools, due to limitations in any single source. Significant effort was required to clean and unify the datasets, especially in standardizing company names and handling inconsistent formats across sources.

After preprocessing, key financial features such as returns, volatility, and rolling metrics were computed. These features allowed a deeper analysis of market behavior, showing clear differences in performance and risk across companies and sectors.

The results showed that some sectors, such as Healthcare, Technology, and Telecom, performed better in terms of growth, while others, such as Basic Materials and Consumer Goods, showed weaker performance. Risk analysis also revealed that volatility varies widely across companies, and that larger companies are not always less risky.

Portfolio simulation demonstrated that sector selection has a strong impact on investment outcomes, with different sectors producing different growth patterns over time.

Finally, network analysis showed that companies are strongly clustered by sector, confirming that industry classification plays an important role in market structure. However, no cross-sector connections were observed in the constructed network model.

this project started with several challenges, including issues in collecting reliable financial data, matching company names across different sources, and understanding how financial metrics like returns and volatility are calculated. A lot of time was spent fixing data problems, cleaning inconsistent formats, and making sure all datasets could be properly merged.

From there, the focus shifted to building meaningful financial analysis, including performance measurement, risk analysis, and portfolio simulation. Through this process, the concepts of financial data analysis became clearer and more practical, especially how market behavior can be measured using data science tools.

Overall, the project evolved from solving technical data issues to building a complete analytical workflow, which improved both programming skills and understanding of financial markets.

The End